

Coding & Solution

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution Solution Summary

Field	Details
Repository Link / URL	https://github.com/nageswari196/credit_card_approval_prediction
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask
Key Features Implemented	• Data preprocessing and cleaning • Feature engineering • Label encoding • SMOTE for balancing the dataset • Model training using Logistic Regression, Decision Tree, Random Forest, and XGBoost • Model evaluation and comparison • Random Forest model selection • Credit card approval prediction • Prediction confidence score • Responsive Flask web application • GitHub repository
Pending / Incomplete Features	• User authentication • Database integration • Explainable AI (SHAP/LIME) • REST API • Email notification • Prediction history storage
Setup / Run Instructions	1. Clone the repository from GitHub. 2. Install dependencies using 'pip install -r requirements.txt'. 3. Run 'python prepare_data.py'. 4. Run 'python train_model.py'. 5. Start the application using 'python app.py'. 6. Open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project is developed using Python and Flask following a modular structure. Separate files are used for data preprocessing (prepare_data.py), model training (train_model.py), and the web application (app.py). Multiple machine learning algorithms were evaluated, and the Random Forest model achieved the best performance with approximately 85.8% accuracy. The application predicts whether a credit card application is likely to be approved or rejected based on applicant information and displays the prediction along with a confidence score. The complete source code is maintained in a GitHub repository.